



Sheriyans

Coding School

BACKEND

YouTube

NOTES



Shreyans



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* NODE JS

→ Export and import in js

- Jab ek chiz export karna hai

```
var a = 12;
```

```
module.exports = a;
```

- Jab do chiz export karna ho

```
var a = 12;
```

```
var b = 12345;
```

```
module.exports = { first: a, second: b }
```

- Import:

```
var data = require("./script");
```

Q) what is node JS?

ans Node js hai khaab saara C++ ka code jo ki liya gaya hai chrome browser ke v8 engine ke code se aur us C++ code jisko liya gaya hai usy wrap kar diya gaya is code se and ye hume allow karta hai ki hum is code likhein aur ye is code accept karega and C++ ke basis par apko ek server environment bana kar dega, par sirf tab tak jab tak ye code chalega.

* Running Node JS code

→ In terminal,

```
node ./script.js
```

name of the file

→ npm: node package manager

npm ek jagah hai jahan khaab saare packages milte hai yani kanoosi features / chizein kehlate hai packages



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- npm = packages ka playstore
package = files of code

* EXPRESS : npm i express

→ Framework for node.js

→ Usecase: Routing

→ Routing $\Rightarrow$ GET, POST, PUT, PATCH, DELETE

→ `const express = require("express")`

`const app = express();`

`app.get("/", function(req, res) {`

`res.send("hello");`

`});`

`app.get("/profile", function(req, res) {`

`res.send("hi");`

`});`

`app.listen(3000);`

- Server ban jata hai after use write express code.
Lekin agar date change hua to humlog ko
uske server restart karna hai.

Is cheez ko hatane ke liye install one package,

`npm i nodemon -g`

`-g`: matlab humesha ~~is~~ rahega jab tak
format na kare.

`g` stands for global.

- nodemon usage

`npmx nodemon ./script.js`



* Middleware: Middleware ek aisa function hai jo kisi bhi route se pehle chalta hai, jiska matlab aapka route chalne se pehle agar aap koi kaam karana chahte ho to ~~to~~ middleware ka upyog kiya jaa sakta hai

```
→ const express = require("express");  
const app = express();  
app.use(function(req, res, next) {  
  console.log("hello from middleware");  
  next();  
});  
app.get("/", function(req, res) {  
  res.send("bye");  
});  
app.listen(3000);
```

→ next kyu likhte hai?

→ sirf ek likhat hai agar humara middle-ware chal gaya to request jam ho jati hai aur route tak nahi pahunch paati, isliye ek push ka yaruwat hota hai jo ki next karta hai taaki route tak jaye.

* Express js:

→ node vs express - node is the main thing express ke code se hum server ka code likh paate and server ka react karega wo bhi express ke help se dikhe hai

→ what is express js - package, routing

→ why express js - http is difficult to use, express make this easier.

→ routing - routes banane ke process ko hum kehte hai routing

→ req, res, next

request mein saara data hota hai jisme user ki request ki taraf ka, jaise uski location, device info and other things, res mein controls hote hai jinke basis pe hum server se response bhej paate hai, next is just a push so that our request moves to the next thing which should be executed.

→ Dynamic routing

facebook.com/profile/harsh

facebook.com/profile/harshit

uska koi bhi route jiska koi hissa baar baar same rehta hai and kuch hissa baar baar change hota hai uske liye ap ek dynamic route bana sakte ho

→

```
app.get("/profile/:username", function(req, res) {  
  res.send('Hi from ' + req.params.username);  
});
```




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→ Template engine $\Rightarrow$ ejs is very similar to HTML

→ ejs setup karne ke liye,

1) ejs install
npm i ejs

2) configure ejs
app.set("view engine", "ejs");

3) ek views folder banao

4) usme ejs files banao

5) uske andar render karke $\neq$ render
karke uske make sure apke views folder
ke andar uske hi koi file ka naam
na ho, aur render fn me ejs mention
na karein

→ script.js
app.get("/", function(req, res) {
 res.render("index", {age: 12});
});

index.ejs

<body>

<div>

<h3> <%= age %> </h3>

</div>

</body>

→ Static files: images, stylesheets, frontend js setup karna
static files setup karne ke liye:

i) Create a folder called public

ii) Create 3 folders inside it, img, stylesheets, javascripts

iii) configure the express static in script.js file

iv) Understand the path



```
→ const express = require("express");  
const app = express();  
app.set("view engine", "ejs");  
app.use(express.static("./public"))  
app.get("/", function(req, res) {  
  res.render("index");  
});
```

→ Error handler :

```
→ app.get("/error", function(req, res, next) {  
  throw Error("something went wrong");  
});
```

```
→ app.use(function errorHandler(err, req, res, next) {  
  if (res.headersSent) {  
    return next(err)  
  }  
  res.status(500)  
  res.render('error', {error: err})  
})  
app.listen(3000);
```

* BACKEND THE ENDGAME 1

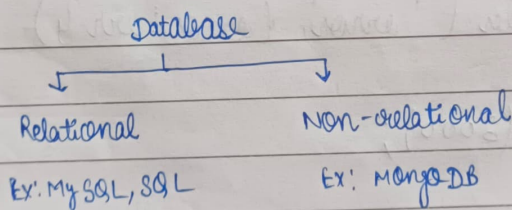
→ Express generator: It'll make the folder structure for you and it'll also write the basic code for the project.



- steps to use express generator:
 - // sabse pehle jeevan me ek saar laptop pe install karke globally.
`npm i express-generator -g`
 - // to create new app anywhere:
open cmd move to desktop
create new app:
`express appname --view = ejs`
 - // now use two commands
`cd appname`
`npm i`
open it on vs code

- pehle aur ab me kya changes hai,
`app.get` → `router.get`
`npm nodemon filename` → `npm nodemon`

* MongoDB:



→ code side
DB setup
model
schema

MongoDB side
DB formation
collection
documents

- Har naye app ka data store hoga storage me, par usse directly rakhne ki jagah ek container me rakhenge, us container me sirf us app ka data rahega.



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→ Schema (code) $\Rightarrow$ documents (dbs)

→ Ek app ka pura data $\Rightarrow$ db

Ek app me variety of data hota hai par pura data hota app ka hi hai, par us data ka subcategory kehlaata hai collection

Collection matlab for ex: users ka data, agar ek user ki last ho rahi to wo document hua.

→ require and setup connection

make schema

create model and export

users.js

```
→ const mongoose = require("mongoose");
mongoose.connect("mongodb://127.0.0.1:27017/practice");
const userSchema = mongoose.schema({
  username: String,
  name: String,
  age: Number
});
mongoose module.exports = mongoose.model("user", userSchema);
```

index.js

```
var express = require("express");
var router = express.Router();
const userModel = require("./users");
router.get("/create", async function(req, res) {
  const createdUser = await userModel.create({
    username: "saran",
    age: 21,
    name: "saran"
  });
  res.send("ban gaya");
});
module.exports = router;
```




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→ router.get("/allusers", async function(req, res) {
 let allusers = await userModel.find();
 res.send(allusers);
});

find: returns an array of all users

→ router.get("/allusers", async function(req, res) {
 let allusers = await userModel.findOne({username: "saranh"});
 res.send(allusers);
});

findOne: Ek user return karaga, based on the condition

→ router.get("/delete", async function(req, res) {
 let deletedUser = await userModel.findOneAndDelete({
 username: "saranh",
 });
 res.send(deletedUser);
});

findOne And Delete: Find karaga & delete

* Cookie and Session

→ Client me cookie hoti hai aur server me session
npm i express-session

app.js

var session = require('express-session')

app.use(session({

 resave: false, //agar session ki value change naa hui
 //to do for all save mat karna

 saveUninitialized: false, //data jisko initialize ni kya gaya
 //hai usko baar baar save ni karna hai

 secret: "abcdeffgh" //secret string jiske madad se data ko
 //encrypt kya jaye
}))



→ Create

```
req.session.Koibhinaam = Koibhivalue;
```

Read

```
req.session.Koibhinaam
```

Delete

```
req.session.destroy
```

```
→ var express = require("express");  
var router = express.Router();  
const userModel = require("./users");  
router.get("/", function (req, res) {  
  req.session.ban = true;  
  res res.render("index", {title: "Express"});  
});  
router.get("/checkboxan", function (req, res) {  
  if (req.session.ban === true) {  
    res.send("you are banned");  
  }  
  else {  
    res.send("not banned");  
  }  
});  
router.get("/removeban", function (req, res) {  
  req.session.destroy(function (err) {  
    if (err) throw err;  
    res.send("ban removed");  
  });  
});
```




→ Cookie setup
res.cookie("name", value);

Cookie reading

req.cookies.name

Cookie delete

res.clearCookie("name")

→ router.get("/", function(req, res) {
 res.cookie("age", 21);
 res.render("index", { title: "Express" });
});

router.get("/read", function(req, res) {
 console.log(req.cookies);
 res.send("check");
});

router.get("/delete", function(req, res) {
 res.clearCookie("age");
 res.send("clear ho gayi");
});



* BACKEND - THE Endgame (PART-2)

* Flash messages :

Jab abhi aap kisi js page ko dekhenge ushan par aapko kisi prakar ka koi information dena hai, wo kehata hai flash messages, they are more like good looking alerts, warning and descriptions.

→ install connect-flash

make sure you setup express-session

make sure you put connect flash in a app.use function

kisi abhi route me aapko flash create karna hai kisi abhi usse route par aap use chlane ka try karen

AAP CONNECT FLASH KO USE NAHI KAR SAKTE BINA EXPRESS-SESSION

→ maye express generator app banane ke bad npm i express-session

app.js

```
var const expresssession = require("express-session")  
const flash = require("connect-flash")  
app.set('view engine', 'ejs')
```

```
app.use(expresssession({
```

```
  resave: false,
```

```
  saveUninitialized: false,
```

```
  secret: "hello669"
```

```
}))
```

```
app.use(flash());
```




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In routes index.js,

→ hum log chahkte hai ki jab hum kisi route par jaye jaise ki /failed us route par ek flash message banne aur wo flash message hum logo ko / route par dikhe jis me

→ flash message ka matlab server ke kisi route me koi data banana and us data ko duere route par peana

→ router.get('/failed', function(req, res) {
 req.flash("age", 12);
 res.send("bangaya");
});

router.get('/checkkaro', function(req, res) {
 console.log(req.flash("age")); // [12]
 res.send("terminal pe check karo");
});

q) How can I perform case-insensitive search in mongoose?

//mongoose ko setup karo by, npm i mongoose

Routes folder ke users.js me,

```
const mongoose = require("mongoose");
```

```
mongoose.connect("mongodb://127.0.0.1:27017/testing");
```

```
const userschema = mongoose.schema({
```

```
  username: String,
```

```
  nickname: String,
```

```
  description: String,
```

```
  categories: {
```

```
    type: Array,
```

```
    default: [] };
```


datacreated: {
 type: Date,
 default: Date.now()
}

3)
 module.exports = mongoose.model("user", userschema);

→ Router folder ke index.js me,

```
var express = require('express');  
var router = express.Router();  
const userModel = require("../user");  
router.get('/', function (req, res) {  
  res.render('index');
```

});

```
router.get("/create", async function (req, res) {  
  let userdata = await userModel.create({  
    username: "harsh",  
    nickname: "async js",  
    description: "I am 21 year old",  
    categories: ['js', 'node', 'react', 'jsdp']  
  });
```

```
  res.send(userdata);
```

});

```
router.get("/find", async function (req, res) {  
  var regex = new RegExp("Harsh$", 'i'); // insensitive (flag)  
  let user = await userModel.find({username: regex});  
  res.send(user);
```

});

module.exports = router;

ans 1)

^ shuruwat
aisi ho

\$ = ant
aisi ho



Q2) How do I find documents where an array field contains all of a set of values?

ans

```
router.get("/find", async function(req, res) {  
  let user = await userModel.find({ categories: { $all: [ "fashion", "science" ] } });  
  res.send(user);  
});
```

Q3) How can I search for documents with a specific date range in mongoose?

ans

```
router.get("/find", async function(req, res) {  
  var date1 = new Date('2023-10-02');  
  var date2 = new Date('2023-10-03');  
  let user = await userModel.find({ datecreated: { $gte: date1, $lte: date2 } });  
  res.send(user);  
});
```

Q4) How can I filter documents based on the existence of a field in mongoose?

ans

```
router.get("/find", async function(req, res) {  
  let user = await userModel.find({ categories: { $exists: true } });  
  res.send(user);  
});
```




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Q) How can I filter documents based on a specific field's length in Mongoose?

ans

```
router.get("/find", async function (req, res) {
  let user = await userModel.find({
    $expr: {
      $and: [
        { $gte: [{ $strlenCP: '$nickname' }, 0] },
        { $lte: [{ $strlenCP: '$nickname' }, 12] }
      ]
    }
  });
  res.send(user);
});
```

* Authentication and Authorization

- Passport me humlog alag alag strategy rakh sakte hai jaise google, local etc.
- passport-local-mongoose package hash karne me help krta hai
- Install these packages,
npm i passport passport-local passport-local-mongoose mongoose express-session
write app.js code first in app.js file and write it after view engine and before logger
Setup users.js then properly
In index.js try register first and then other codes as well



app.js

```
→ app.use(expressSession({
  resave: false,
  saveUninitialized: false,
  secret: "hello"
}));
app.use(passport.initialize());
app.use(passport.session());
passport.serializeUser(usersRouter.serializeUser());
passport.deserializeUser(usersRouter.deserializeUser());
```

users.js

```
const mongoose = require("mongoose");
const fplm = require("passport-local-mongoose");
mongoose.connect("mongodb://127.0.0.1:27017/test");
const userSchema = mongoose.Schema({
  username: String,
  password: String,
  secret: String
});
userSchema.plugin(fplm);
module.exports = mongoose.model("user", userSchema);
```

index.js

```
const localStrategy = require("passport-local");
passport.use(new localStrategy(userModel.authenticate()));
router.get('/', function(req, res) {
  res.render('index');
});
router.post('/register', function(req, res) {
  var userdata = new userModel({
    username: req.body.username,
    secret: req.body.secret
  });
```



```

Protected Route {
  router.get('/profile', isLoggedIn, function (req, res) {
    res.send('welcome to profile');
  });
  userModel.register(userData, req.body.password)
    .then(function (registeredUser) {
      passport.authenticate("local")(req, res, function () {
        res.redirect('/profile');
      });
    });
});

router.post("/login", passport.authenticate("local", {
  successRedirect: "/profile",
  failureRedirect: "/"
}), function (req, res) {});

router.get("/logout", function (req, res, next) {
  req.logout(function (err) {
    if (err) return next(err);
    res.redirect("/");
  });
});

function isLoggedIn (req, res, next) {
  if (req.isAuthenticated()) {
    return next();
  }
  res.redirect("/");
}

module.exports = router;

```